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An improved time-varying mesh stiffness algorithm and dynamic modeling of gear-rotor system with tooth root crack

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ABSTRACT

When a tooth crack failure occurs, the vibration response characteristics caused by the change of time-varying mesh stiffness play an important role in crack fault diagnosis. In this paper, an improved time-varying mesh stiffness algorithm is presented. A coupled lateral and torsional vibration dynamic model is used to simulate the vibration response of gear-rotor system with tooth crack. The effects of geometric transmission error (GTE), bearing stiffness, and gear mesh stiffness on the dynamic model are analyzed. The simulation results show that the gear dynamic response is periodic impulses due to the periodic sudden change of time varying mesh stiffness. When the cracked tooth comes in contact, the impulse amplitude will increase as a result of reductions of mesh stiffness. Amplitude modulation phenomenon caused by GTE can be found in the simulation signal. The lateral–torsional coupling frequency increases greatly within certain limits and thereafter reaches a constant while the lateral natural frequency nearly remains constant as the gear mesh stiffness increases. Finally, an experiment was conducted on a test bench with 2 mm root crack fault. The results of experiment agree well with those obtained by simulation. The proposed method improves the accuracy of using potential energy method to calculate the time-varying mesh stiffness and expounds the vibration mechanism of gear-rotor system with tooth crack failure.

1. Introduction

Gear-rotor system is one of the most common mechanisms for transmission of motion and power in industry. When machines work under poor working conditions, unexpected failure occurs in gear-rotor system frequently. Hence, condition monitoring and fault diagnosis of gear-rotor systems is an important task for condition-based maintenance of machines. Although, vibration signals are widely used in damage detection of gearboxes, measured signals are often affected by the complicated system coupling and transfer path between the excitation source and accelerometer. Dynamic modeling and vibration simulation are necessary in order to investigate the damage mechanism and characteristics of vibration signals, which can provide theoretical supports for running state identification of gear-rotor systems.

To improve current techniques of gearbox vibration monitoring and diagnosis, many works have been done on the gear dynamic modeling to ascertain the effect of different types of gear damage on the gear vibration. Özgüven and Houser [1] presented a comprehensive review of mathematical models used in gear dynamics. Howard et al. [2] explored the effects of friction and single tooth crack on the gear case vibration by using a simplified gear dynamic model. Bartelmus [3] dealt

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with mathematical modeling and computer simulation as a tool for aiding gearbox diagnostic inference. Parey et al. [4] developed a 6 degrees of freedom gear dynamic model including localized tooth defect. Jia and Howard [5] presented a 26 degrees of freedom gear dynamic model involves three shafts and two pairs of spur gears in mesh for comparison of localized tooth spalling and crack. Ma and Chen [6] used a four degrees of freedom gear dynamic model to study the dynamic and vibration characteristics of gear system with local defects. Randall, Sawalhi and Endo [7,8] used the simulated fault signals to design and improve the fault detection and machine condition monitoring techniques.

In all of the gear dynamic models mentioned above, the time varying mesh stiffness is one of the most important excitation sources of vibration. The most widely used technique to get time varying mesh stiffness is finite element method, but it requires mesh refinements and more computations time [9]. The software ROMAX can also be used to calculate mesh stiffness, but it is unable to calculate the mesh stiffness when the tooth crack occurs [10]. The ISO standard 6336-1-2006 defines a method to calculate the mesh stiffness, but it can only calculate the single stiffness and average mesh stiffness. A so-called potential energy method which was presented by Yang and Lin [11] can calculate the time varying stiffness conveniently and effectively. Tian [12] refined the model by taking shear energy into account. Wu et al. [13] developed the method when introducing straight line crack growth path into the model. Based on this method, Pandya and Parey [14,15] studied the effect of crack path on mesh stiffness under different gear parameters like contact ratio, pressure angle, fillet radius and backup ratio. But they did not take the fillet-foundation deflections and the flexibility between base circle and root circle into account.

For a gearbox system, the coupled phenomenon of lateral and torsional vibrations cannot be ignored [16]. In most of the gear dynamic model mentioned above, they assumed the shaft as rigid or used the equivalent torsional stiffness to represent the real torsional stiffness. However, this method is not very strict obviously, it may not fully account for the shaft-bearing structural characteristics accurately.

In this study, a more accurate mathematical model which takes the fillet-foundation deflections and the flexibility between root circle and the base circle into account is presented in the calculation of time varying stiffness. A coupled lateral and torsional dynamic model is used to simulate the fault vibration response of gear-rotor system. The effects of GTE, bearing stiffness, and gear mesh stiffness on the dynamic model are analyzed. Then an experiment was conducted on a test bench with root crack fault. Finally, by comparing the simulation results with the experiments results, the validity of the model is verified, and the principle of gear root crack fault is analyzed.

2. Calculation of time varying mesh stiffness

In gear pair systems, the coupling between the torsional vibration and lateral vibration is mainly controlled by time varying mesh stiffness, and thus it is very important to calculate the mesh stiffness conveniently and effectively.

In the so-called energy method [13], the total potential energy stored in the meshing gear system was assumed to include four components: Hertzian energy U_h, bending energy U_b, shear energy U_s and axial compressive energy U_a which can be used to calculate Hertzian mesh stiffness k_h, bending mesh stiffness k_b, shear mesh stiffness k_s and axial compressive stiffness k_a, respectively. According to material mechanics and elastic mechanics, U_h, U_b, U_s, U_a can be expressed as follows:

$$U_h = \frac{F^2}{2k_h} \quad \text{where } K_h = \frac{\pi EL}{4(1-v^2)} \tag{1}$$

$$U_b = \frac{F^2}{2k_b} = \int_0^d \frac{[F_b(d-x) - F_a h]^2}{2EI_x}\,\mathrm{d}x \tag{2}$$

$$U_s = \frac{F^2}{2k_s} = \int_0^d \frac{1.2F_b^2}{2GA_x}\,\mathrm{d}x \tag{3}$$

$$U_a = \frac{F^2}{2k_a} = \int_0^d \frac{F_a^2}{2EA_x}\,\mathrm{d}x \tag{4}$$

$$I_x = \frac{1}{12}(2h_x)^3 L \tag{5}$$

$$A_x = (2h_x)L \tag{6}$$

where F represents the acting force by the mating tooth in the contact point. F_a, F_b are radial and tangential forces, G, E, L, v represent shear modulus, Young's modulus, the width of tooth and Poisson's ratio, respectively. I_x, A_x are the area moment of inertia and area of the section where the distance from the tooth root is x, the other parameters are shown in Fig. 1.

When two gears are meshing, the total energy can be obtained by

$$U = \frac{F^2}{2k} = U_h + U_{b1} + U_{s1} + U_{a1} + U_{b2} + U_{s2} + U_{a2} = \frac{F^2}{2}\left(\frac{1}{k_h} + \frac{1}{k_{b1}} + \frac{1}{k_{s1}} + \frac{1}{k_{a1}} + \frac{1}{k_{b2}} + \frac{1}{k_{s2}} + \frac{1}{k_{a2}}\right) \tag{7}$$

where k represents the total effective mesh stiffness, subscripts 1 and 2 indicate the driving and driven gears, respectively.

Beside the tooth deformation, the fillet-foundation deflection which can be computed by using the theory of Muskhelishvili [17] also influences the stiffness of gear tooth [18,19], and it can be calculated by

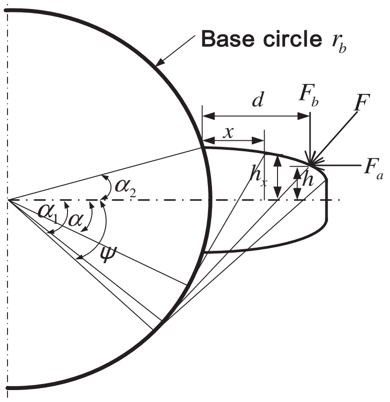


Fig. 1. Model of spur gear tooth.

$$\frac{1}{k_f} = \frac{\cos^2 \alpha_m}{EL} \left\{ L^* \left(\frac{\mu_f}{S_f} \right)^2 + M^* \left(\frac{\mu_f}{S_f} \right) + P^* \left(1 + Q^* \tan^2 \alpha_m \right) \right\} \tag{8}$$

The coefficients L^*, M^*, P^*, Q^* can be approached by polynomial functions

$$X_i^* = A_i/\theta_f^2 + B_i h_{fi}^2 + C_i h_{fi}/\theta_f + E_i h_{fi} + F_i \tag{9}$$

the values of A_i, B_i, C_i, D_i, E_i, F_i are given in Table 1. L is the tooth width u_f, s_f $h_{fi} = r_f/r_{\text{int}}$ and θ_f are defined in Fig. 2.

In conclusion, for gear teeth in contact, the total effective mesh stiffness can be written as:

$$k = \begin{cases} 1\Big/\left(\frac{1}{k_h} + \frac{1}{k_{b1}} + \frac{1}{k_{s1}} + \frac{1}{k_{f1}} + \frac{1}{k_{a1}} + \frac{1}{k_{b2}} + \frac{1}{k_{s2}} + \frac{1}{k_{a2}} + \frac{1}{k_{f2}}\right) & \text{single-tooth-pair} \\ \sum_{i=1}^{2} \frac{1}{\frac{1}{k_{h,i}} + \frac{1}{k_{b1,i}} + \frac{1}{k_{s1,i}} + \frac{1}{k_{f1,i}} + \frac{1}{k_{a1,i}} + \frac{1}{k_{b2,i}} + \frac{1}{k_{s2,i}} + \frac{1}{k_{a2,i}} + \frac{1}{k_{f2,i}}} & \text{double-tooth-pair} \end{cases} \tag{10}$$

where subscripts 1 and 2 denote the driving and driven gears respectively, $i = 1$ represents the first pair of meshing teeth, $i = 2$ stands for the second pair of meshing teeth.

This so called potential energy method is time-saving comparing with ANSYS method. It is suitable to calculate the meshing stiffness when there is a crack fault in the gear pair [10].

2.1. The improved energy method

In spite of those advantages of this method, it has a defect which needs to be corrected as follows.

Before indicating the defect, a principle of mechanics about gears needs to be known. According to the knowledge of principle of mechanics, the radius of gear base circle r_b and root circle r_f can be calculated as:

$$r_b = \frac{mz}{2}\cos(\theta), \quad r_f = \frac{mz}{2} - (h_a^* + c^*)m \tag{11}$$

where m, z, θ represent module, number of teeth and pressure angle, respectively. h_a^*, c^* are addendum and tip clearance coefficients. If $h_a^* = 1$, $c^* = 0.25$, $\theta = 20^\circ$, when r_b is equal to r_f, the number of teeth $z \approx 42$ can be obtained by using formula (11).

The potential energy method assumes the gear tooth as a variable cross-section cantilever beam which is fixed on the gear base circle. In this way, the potential energy stored in the part between base circle and root circle (the grid line portions as shown in Fig. 3) will be ignored. This defect must be corrected for further accurate calculation of time varying mesh stiffness.

Table 1
Values of the coefficients of Eq. (9).

	A_i	B_i	C_i	D_i	E_i	F_i
L^*	-5.574×10^{-5}	-1.9986×10^{-3}	-2.3015×10^{-4}	4.7702×10^{-3}	0.0271	6.8045
M^*	60.111×10^{-5}	-28.100×10^{-3}	-83.431×10^{-4}	-9.9256×10^{-3}	0.1624	0.9086
P^*	-50.952×10^{-5}	185.50×10^{-3}	0.0538×10^{-4}	53.3×10^{-3}	0.2895	0.9236
Q^*	-6.2042×10^{-5}	9.0889×10^{-3}	-4.0964×10^{-4}	7.8297×10^{-3}	-0.1472	0.6904

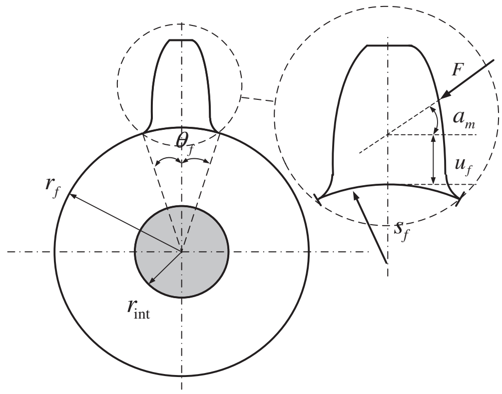


Fig. 2. Geometrical parameters for the fillet-foundation deflection.

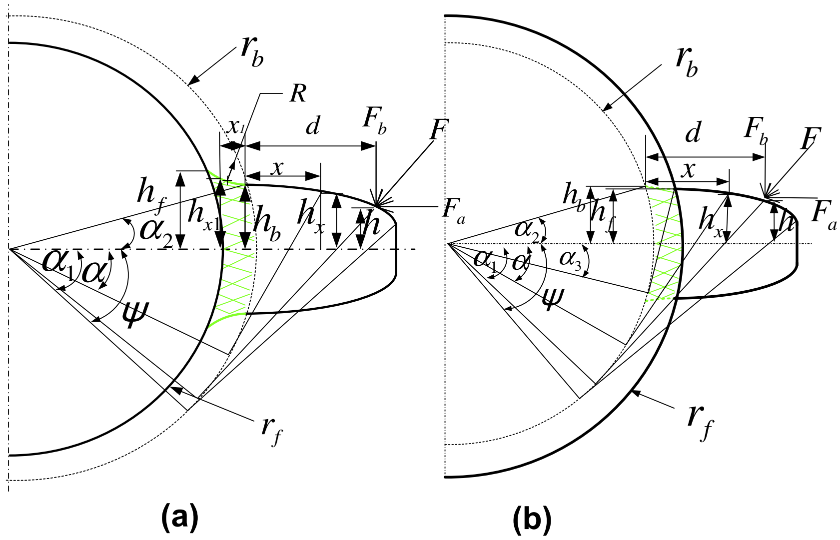


Fig. 3. Schematic of spur gear when $z < 42$ (a) and $z > 42$ (b).

If the number of teeth is less than 42 (see in Fig. 3(a)), $r_b > r_f$ the energy stored in the part between base and root circles must be added on the original foundation. If the number of teeth is more than 42 (see in Fig. 3(b)), the energy stored in the part between base circle and root circle must be subtracted on the original foundation.

Therefore, when the number of teeth is less than 42, the formula of bending potential energy U_b (see in formula (2)) can be corrected as

$$U_b = \int_0^d \frac{[F_b(d-x) - F_a h]^2}{2EI_x} dx +_1 \int_0^{r_b - r_f} \frac{[F_b(d+x_1) - F_a h]^2}{2EI_{x_1}} dx_1 \tag{12}$$

where I_{x_1} represents the area moment of inertia of the section where the distance from the tooth root is x_1. Form the geometry of involute profile, d, x, h_x can be expressed as

$$\begin{cases} d = r_b[\cos\alpha_1 + (\alpha_1 + \alpha_2)\sin\alpha_1 - \cos\alpha_2] \\ x = r_b[\cos\alpha - (\alpha_2 - \alpha)\sin\alpha - \cos\alpha_2] \\ h_x = r_b[(\alpha_2 + \alpha)\cos\alpha - \sin\alpha] \end{cases} \tag{13}$$

After simplify the formula (12), the bending mesh stiffness K_b can be expressed as

$$k_b = 1 \bigg/ \left(\int_{-\alpha_1}^{\alpha_2} \frac{3\{1 + \cos\alpha_1[(\alpha_2 - \alpha_1)\sin\alpha - \cos\alpha]\}^2 (\alpha_2 - a)\cos\alpha}{2EL[\sin\alpha + (\alpha_2 - \alpha)\cos\alpha]^3} \mathrm{d}\alpha + \int_0^{r_b - r_f} \frac{[(d + x_1)\cos\alpha_1 - h\sin\alpha_1]^2}{EI_{x_1}} \mathrm{d}x_1 \right) \tag{14}$$

where h_{x_1} is the distance between the point on the tooth curve and the central line of the tooth, it can be expressed as $h_{x_1} = h_b + R - \sqrt{R^2 - x_1^2}$, $h_b = r_b \sin(\alpha_2)$, R is root fillet radius.

Similarly, the shear mesh stiffness k_s and axial compressive stiffness k_a will be

$$k_s = 1 \Big/ \left(\int_{-\alpha_1}^{\alpha_2} \frac{1.2(1+v)(\alpha_2 - a)\cos\alpha\cos^2\alpha_1}{EL[\sin\alpha + (\alpha_2 - \alpha)\cos\alpha]} d\alpha + \int_0^{r_b - r_f} \frac{(1.2\cos\alpha_1)^2}{GA_{x_1}} dx_1 \right) \tag{15}$$

$$k_a = 1 \Big/ \left(\int_{-\alpha_1}^{\alpha_2} \frac{(\alpha_2 - a)\cos\alpha\sin^2\alpha_1}{2EL[\sin\alpha + (\alpha_2 - \alpha)\cos\alpha]} d\alpha + \int_0^{r_b - r_f} \frac{(\sin\alpha_1)^2}{EA_{x_1}} dx_1 \right) \tag{16}$$

where A_{x_1} represents the area of the section where the distance from the tooth root is x_1.

When the number of teeth is more than 42, the formula of bending potential energy U_b (see in formula (2)) can be corrected as

$$U_b = \int_{r_b - r_f}^{d} \frac{[F_b(d-x) - F_a h]^2}{2EI_x} dx \tag{17}$$

The formula of bending mesh stiffness K_b can be corrected as:

$$k_b = 1 \Big/ \left(\int_{-\alpha_1}^{-\alpha_3} \frac{3(\alpha_2 - a)\cos\alpha\{1 + \cos\alpha_1[(\alpha_2 - \alpha_1)\sin\alpha - \cos\alpha]\}^2 d\alpha}{2EL[\sin\alpha + (\alpha_2 - \alpha)\cos\alpha]^3} \right) \tag{18}$$

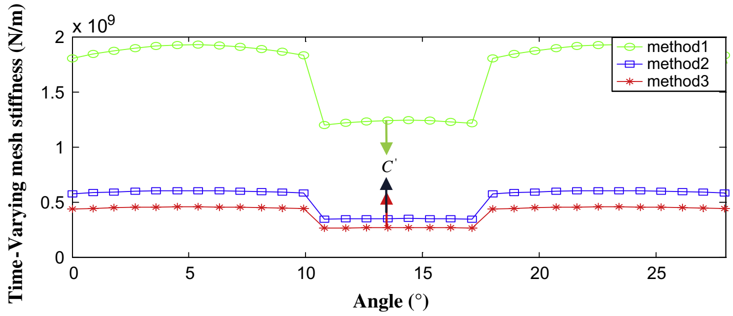
Similarly, the shear mesh stiffness k_s and axial compressive stiffness k_a will be

$$k_s = 1 \Big/ \left(\int_{-\alpha_1}^{-\alpha_3} \frac{1.2(1+v)(\alpha_2 - a)\cos\alpha\cos^2\alpha_1}{EL[\sin\alpha + (\alpha_2 - \alpha)\cos\alpha]} d\alpha \right) \tag{19}$$

$$k_a = 1 \Big/ \left(\int_{-\alpha_1}^{-\alpha_3} \frac{(\alpha_2 - a)\cos\alpha\sin^2\alpha_1}{2EL[\sin\alpha + (\alpha_2 - \alpha)\cos\alpha]} d\alpha \right) \tag{20}$$

where $\alpha_3 = \alpha_f - \left(\frac{\pi}{2z} + (\theta - \theta_f)\right)$, $\theta = inv(\alpha)$, $\theta_f = inv(\alpha_f)$. α and α_f are the pressure angle of reference circle and root circle, respectively.

Now, in order to confirm the accuracy of the improved time-varying mesh stiffness algorithm, three methods will be compared with ISO standard 6336-1-2006 which can be utilized to calculate the single stiffness c' and mesh stiffness c_γ. As



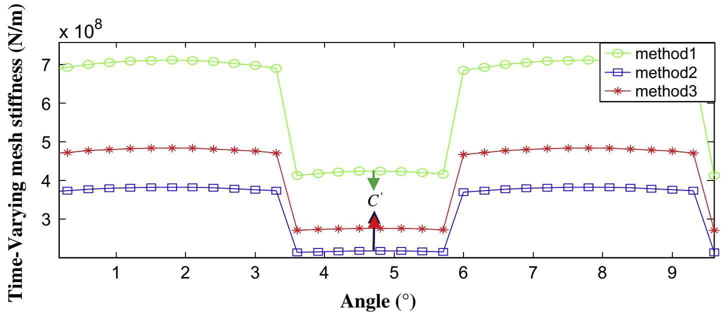


Fig. 4. The comparison of time-varying mesh stiffness calculated by three methods (a) $z = 20$ and (b) $z = 60$.

Table 2
The relative error of three methods compared with ISO standard when the number of teeth z is equal to 20.

	c' ($\times 10^8$ (N/m))	Relative error (%)	c_γ ($\times 10^8$ (N/m))	Relative error (%)
ISO	2.34	0	3.4075	0
Method 1	12.45	432	15.89	366
Method 2	3.51	50	4.72	38.5
Method 3	2.69	14	3.488	2.4

Table 3
The relative error of the three methods compared with ISO standard when the number of teeth z is equal to 60.

	c' ($\times 10^8$ (N/m))	Relative error (%)	c_γ ($\times 10^8$ (N/m))	Relative error (%)
ISO	2.955	0	4.284	0
Method 1	4.23	52.3	5.70	35.7
Method 2	2.17	−26.6	3.02	−29.5
Method 3	2.75	−6.9	3.95	−7.7

shown in Fig. 4, Tables 2 and 3, the three methods are referred as method 1, method 2 and method 3. The method 1 was proposed by Wu et al., [13], the method 2 based on the method 1 was developed by introducing the fillet-foundation deflection, the improved method of this paper is named the method 3.

ISO standard 6336-1-2006 defined c' as the maximum stiffness of a single pair of spur gear teeth and c_γ as the mean value of stiffness of all the teeth in a mesh. c' can be expressed as

$$c' = c'_{th} C_M C_R C_B \cos(\beta) \tag{21}$$

where Correction factor C_M = 0.8, Gear blank factor C_R = 1, Basic rack factor C_B = 1, β is helical angle, theoretical single stiffness $c'_{th} = \frac{1}{q'}$, q' can be calculated as

$$q' = C_1 + \frac{C_2}{z_{n1}} + \frac{C_3}{z_{n2}} + C_4 x_1 + \frac{C_5 x_1}{z_{n1}} + C_6 x_2 + \frac{C_7 x_2}{z_{n2}} + C_8 x_1^2 + C_9 x_2^2 \tag{22}$$

See Table 4 for coefficients C_1–C_9

Mesh stiffness c_γ is

$$c_\gamma = (0.75\varepsilon_\alpha + 0.25)c' \tag{23}$$

where ε_α is contact ratio factor.

Tables 2 and 3 show the comparison of c' and c_γ calculated by the three methods. Fig. 4 is the comparison of time varying stiffness calculated by the three methods.

As shown in Fig. 4(a) and Table 2, when the number of teeth is equal to 20, the result by using method 1 is relatively greater, the relative error of c' reaches to 432%. Since the flexibility between the root circle and tooth base circle is not considered, the relative error of c' by using the method 2 also reaches to 50%.

When the number of teeth is 60, Fig. 4(b) and Table 3 show that the relative error of c' by using method 2 is 26.6% less than the result by ISO standard, but the relative error of c' by using method 3 is only 6.9%.

In conclusion, the method 3 proposed in this paper is most accurate for calculating time varying mesh stiffness.

2.2. Calculation of mesh stiffness of gear with tooth root crack

A tooth root crack typically starts at the point of the largest stress in the material. Lewicki [20] indicates that crack propagation paths are smooth, continuous, and in most cases, rather straight with only a slight curvature. This paper also assumes the cracked tooth as a cantilevered beam, the intersection angle v between the crack and the central line of the tooth is a constant (see in Fig. 5), and the curve of the tooth profile remain perfect. In this case the Hertzian contact stiffness k_h and axial compressive stiffness k_a will not change at all, and they also can be calculated according to Eqs. (1), (16) and (20).

However, the bending and shear stiffness will change due to the influence of the crack. When the crack is present, the effective area moment of inertia and area of the cross section at a distance of x from the tooth root will be calculated as Eqs. (24) and (25), respectively. In the same way, form the geometry of involute profile, after simplification, the bending k_b stiffness and shear stiffness k_s can be obtained.

Table 4
Coefficients for Eq. (22).

C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
0.04723	0.15551	0.25791	−0.00635	−0.11654	−0.00193	−0.24188	0.00529	0.00182

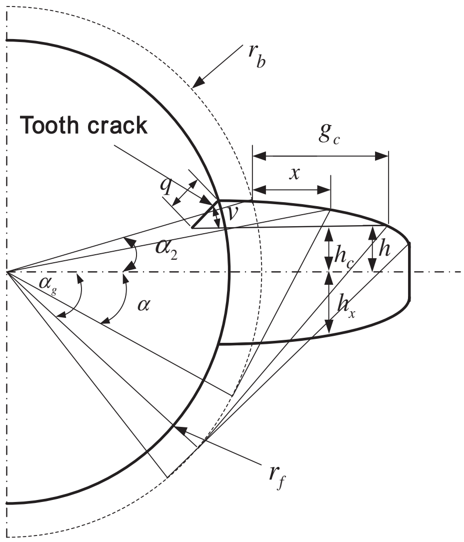


Fig. 5. A crack tooth model.

$$I_x = \begin{cases} \frac{1}{12}(h_c + h_x)^3 L & x \leqslant g_c \\ \frac{1}{12}(2h_x)^3 L & x > g_c \end{cases} \tag{24}$$

$$A_x = \begin{cases} (h_c + h_x)L & x \leqslant g_c \\ (2h_x)L & x > g_c \end{cases} \tag{25}$$

where h_c is the distance from the root of the crack to the central of the tooth.

An example of a single-stage spur gear system is presented in Table 5. The gear mesh stiffness of the gear pair with tooth crack calculated according to Eq. (10) is shown in Fig. 6. It shows that the mesh stiffness will reduce when the cracked tooth on pinion comes into contact with the gear during meshing. The total effective mesh stiffness reduces further with increase in crack length.

3. The coupled lateral and torsional vibrations model of a gear-rotor system

As shown in Fig. 7(a), a typical gear-rotor system is usually composed of gear pair, bearing, elastic shaft, drive motor and so on. Fig. 7(b) shows the dynamic model of the gear-rotor system. The dynamic modeling process of the gear-rotor system can be described as follows. Firstly, the gear-rotor system can be decomposed into gear pair system and rotor-bearing system. Secondly, the rotor-bearing system can be modeled by the finite element method, the stiffness, mass matrix and gyroscopic matrix of beam element are derived and assembled to form the total mass matrix $\mathbf{M}_R$, stiffness matrix $\mathbf{K}_R$ and the gyroscopic matrix $\mathbf{G}_R$ of rotor-bearing system. The gear pair system can be analyzed by typical mathematical model developed by Lee and Ha [21]. Similarly, the mass matrix $\mathbf{M}_G$, stiffness matrix $\mathbf{K}_G$, gyroscopic matrix $\mathbf{G}_G$ and damping matrix $\mathbf{C}_G$ of the gear pair system are derived from the free vibration equation, this step will be introduced in detail in Section 3.2. And

Table 5
The parameters of spur gear rig.

	Driving gear	Driven gear
Number of tooth	55	75
Module (mm)	2	2
Pressure angle (°)	20	20
Young's modulus (Pa)	2.06×10^{11}	2.06×10^{11}
Poisson's ratio	0.3	0.3
With of tooth (mm)	20	20
Moment of inertia (kg m^2)	0.0028	0.0097
Eccentricity (um)	60	60
Stiffness of bearing (N/m)	2×10^8	
Amplitude of tooth profile error (um)	10	
Phase difference (rad)	0	

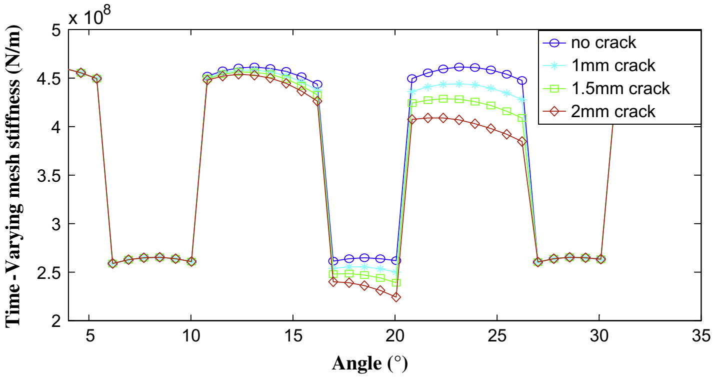


Fig. 6. Mesh stiffness for different crack size.

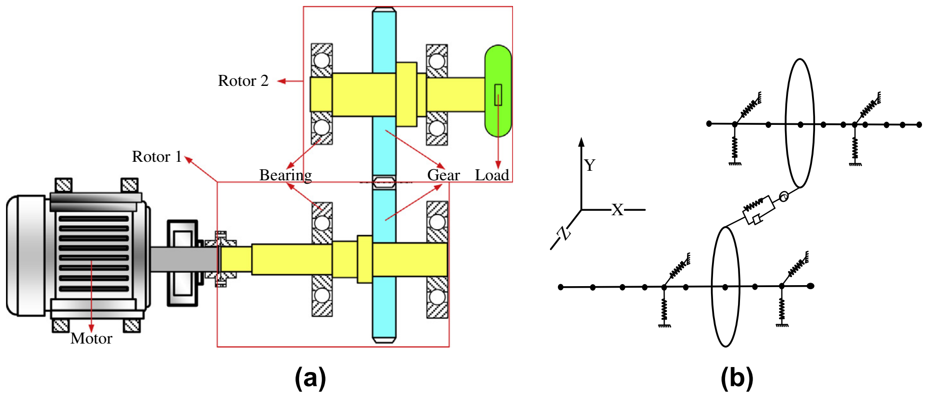


Fig. 7. (a) Typical gear-rotor system configuration and (b) schematic of setup for the dynamic finite element modeling of gear-rotor system.

lastly, the total matrix $\mathbf{M}_{total}$, $\mathbf{K}_{total}$, $\mathbf{G}_{toatal}$, $\mathbf{C}_{toatal}$ can be obtained by assembling the matrices $\mathbf{M}_G$, $\mathbf{K}_G$, $\mathbf{G}_G$, $\mathbf{C}_G$ of gear pair system onto the matrices of the rotor-bearing system. Thus the vibration differential equation of gear-rotor system is deduced, it can be shown as follows.

$$\mathbf{M}_{total}\ddot{\mathbf{q}} + (\mathbf{G}_{total} + \mathbf{C}_{total})\dot{\mathbf{q}} + \mathbf{K}_{total}\mathbf{q} = \mathbf{Q}_{total} \tag{26}$$

The displacement vector $\mathbf{q}$ in formula (26) is generalized displacements of each node and the force vector $\mathbf{Q}_{total}$ is generalized force.

3.1. Modeling of rotor-bearing system

The typical flexible rotor-bearing system to be analyzed consists of a rotor composed of discrete disks, rotor segments with distributed mass and elasticity, and discrete bearings. Such a system is illustrated in Fig. 8.

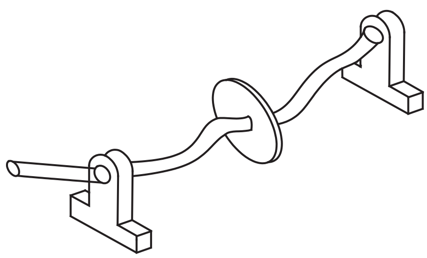


Fig. 8. Typical rotor-bearing system configuration.

3.1.1. The dynamic model of elastic axis

According to the principle of finite element method, dividing the rotor into a number of beam elements (see in Fig. 9(a)). The stiffness matrix, mass matrix and the gyroscopic matrix of two-node beam elements is shown in Fig. 3(b) are deduced [22]. Each node of the beam elements has six degrees of freedom x, y, z are displacements in X–Y–Z, and θ_x, θ_y, θ_z are rotation angles of the three directions, respectively. The node displacements of the rotor element can be expressed as

$$\mathbf{q}^e = [x_A \; y_A \; z_A \; \theta_{Ax} \; \theta_{Ay} \; \theta_{Az} \; x_B \; y_B \; z_B \; \theta_{Bx} \; \theta_{By} \; \theta_{Bz}]^T \tag{27}$$

The energy of the complete element is obtained by summing the elastic bending and kinetic energy expressions together.

$$E = \frac{1}{2}\dot{\mathbf{q}}^{e^T}(\mathbf{M}_T^e + \mathbf{M}_R^e)\dot{\mathbf{q}}^e + \frac{1}{2}J_p\omega^2 + \omega\dot{\mathbf{q}}^{e^T}\mathbf{G}^e\mathbf{q}^e + \frac{1}{2}\mathbf{q}^{e^T}\mathbf{K}^e\mathbf{q}^e \tag{28}$$

The Lagrange equation of motion for the finite rotor element using Eq. (28) is

$$(\mathbf{M}_T^e + \mathbf{M}_R^e)\ddot{\mathbf{q}}^e - \omega\mathbf{G}^e\dot{\mathbf{q}}^e + \mathbf{K}^e\mathbf{q}^e = \mathbf{Q}^e \tag{29}$$

All the matrices of Eq. (29) presented in Appendix A are symmetric except the gyroscopic matrix $\mathbf{G}^e$ which is skew symmetric, they are given in Appendix A by Eqs. (A.1)–(A.4). According to relative principle, the element matrix can be assembled to total matrix $\mathbf{M}_R$, stiffness matrix $\mathbf{K}_R$ and gyroscopic matrix $\mathbf{G}_R$ of rotor.

3.1.2. The dynamic model of discrete bearings

The bearings utilized in this paper can be simplified as a spring-damper system as shown in Fig. 10. The central coordinate of the bearing support is $[x_b \; y_b]$, and shaft neck is $[x_s \; y_s]$, the vibration differential equation of the bearing support can be expressed as

$$\begin{bmatrix} M_{bx} & 0 \\ 0 & M_{by} \end{bmatrix}\begin{Bmatrix} \ddot{x}_b \\ \ddot{y}_b \end{Bmatrix} + \begin{bmatrix} c_{xx} & c_{xy} \\ c_{yx} & c_{yy} \end{bmatrix}\begin{Bmatrix} \dot{x}_b - \dot{x}_s \\ \dot{y}_b - \dot{y}_s \end{Bmatrix} + \begin{bmatrix} k_{xx} & k_{xy} \\ k_{yx} & k_{yy} \end{bmatrix}\begin{Bmatrix} x_b - x_s \\ y_b - y_s \end{Bmatrix} + \begin{bmatrix} c_{bxx} & c_{bxy} \\ c_{byx} & c_{byy} \end{bmatrix}\begin{Bmatrix} \dot{x}_b \\ \dot{y}_b \end{Bmatrix} + \begin{bmatrix} k_{bxx} & k_{bxy} \\ k_{byx} & k_{byy} \end{bmatrix}\begin{Bmatrix} x_b \\ y_b \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{30}$$

If the bearing support is approximated rigid, then $x_b = 0$, $y_b = 0$, then the generalized force of the bearing support acting on the shaft neck can be expressed as

$$\begin{Bmatrix} Q_1 \\ Q_2 \end{Bmatrix} = -\begin{bmatrix} c_{xx} & c_{xy} \\ c_{yx} & c_{yy} \end{bmatrix}\begin{Bmatrix} \dot{x}_s \\ \dot{y}_s \end{Bmatrix} - \begin{bmatrix} k_{xx} & k_{xy} \\ k_{yx} & k_{yy} \end{bmatrix}\begin{Bmatrix} x_s \\ y_s \end{Bmatrix} \tag{31}$$

Further assume that the support is isotropic and no coupling, the formula (31) can then be expressed as

$$\begin{Bmatrix} Q_1 \\ Q_2 \end{Bmatrix} = -\begin{bmatrix} c_{xx} & 0 \\ 0 & c_{yy} \end{bmatrix}\begin{Bmatrix} \dot{x}_s \\ \dot{y}_s \end{Bmatrix} - \begin{bmatrix} k_{xx} & 0 \\ 0 & k_{yy} \end{bmatrix}\begin{Bmatrix} x_s \\ y_s \end{Bmatrix} \tag{32}$$

3.2. Modeling of gear pair system

In recent decades, the gear dynamic analysis has drawn wide attention. As research continues, the gear dynamic model becomes more and more complex [23]. The dynamic model of gear pair system utilized in this paper is ten degrees of freedom nonlinear model [16] which will be introduced as follows.

3.2.1. Nonlinear model of gear pair system

A symbolic representation of a single stage gear system is illustrated in Fig. 11. A pair of meshing gears is modeled by rigid disks representing their mass/moment of inertia. The disks are linked by line elements that represent the stiffness and the damping. As shown in Fig. 11, a displacement vector of a gear pair can be defined from the pressure line co-ordinate system, the central coordinate vector can be expressed as

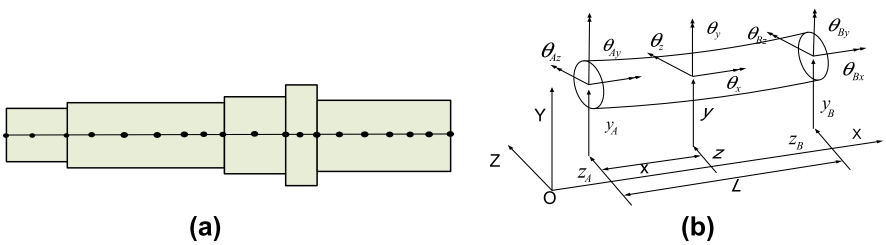


Fig. 9. Typical finite rotor element and coordinates.

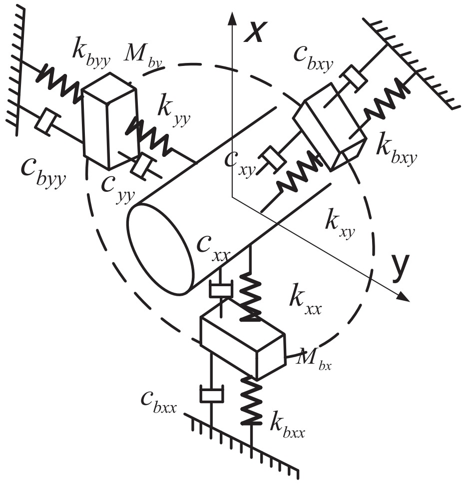


Fig. 10. Dynamic model of bearing.

$$\mathbf{q}_G = [u_1 \ v_1 \ \theta_1 \ \theta_{u1} \ \theta_{v1} \ u_2 \ v_2 \ \theta_2 \ \theta_{u2} \ \theta_{v2}]^T$$

where u, v, θ_u, θ_v are the lateral degrees of freedom and θ_1, θ_2 are the torsional degree of freedom. Subscripts 1 and 2 indicate the driving and driven gears. O_1 and O_2 are centers of the gears when they are stationary, O_1' and O_2' are centers of the gears when they are rotating, and G_1 and G_2 geometrical centers of the gears. From the equilibrium of the force, bending moment and torque for each gear, the equations of motion of a gear pair can be expressed by the following equations:

$$\begin{aligned} &M_1\ddot{\mu}_1 + K_m(\mu_1 + r_1\theta_1 - \mu_2 + r_2\theta_2) + C_m\left(\dot{\mu}_1 + r_1\dot{\theta}_1 - \dot{\mu}_2 + r_2\dot{\theta}_2\right) \\ &\quad = M_1 e_1 \theta_1^2 \sin(\theta_1) + K_m(e_2\sin\theta_2 - e_1\sin\theta_1 + e_t) + C_m(e_2\omega_2\sin\theta_2 - e_1\omega_1\sin\theta_1 + \dot{e}_t) \end{aligned} \tag{33}$$

$$M_1\ddot{v}_1 = M_1 e_1 \theta_1^2 \cos(\theta_1) \tag{34}$$

$$\begin{aligned} &M_2\ddot{\mu}_2 - K_m(\mu_1 + r_1\theta_1 - \mu_2 + r_2\theta_2) - C_m\left(\dot{\mu}_1 + r_1\dot{\theta}_1 - \dot{\mu}_2 + r_2\dot{\theta}_2\right) \\ &\quad = M_1 e_1 \theta_1^2 \sin(\theta_1) - [K_m(e_2\sin\theta_2 - e_1\sin\theta_1 + e_t) + C_m(e_2\omega_2\sin\theta_2 - e_1\omega_1\sin\theta_1 + \dot{e}_t)] \end{aligned} \tag{35}$$

$$M_2\ddot{v}_2 = M_2 e_2 \theta_2^2 \cos(\theta_2) \tag{36}$$

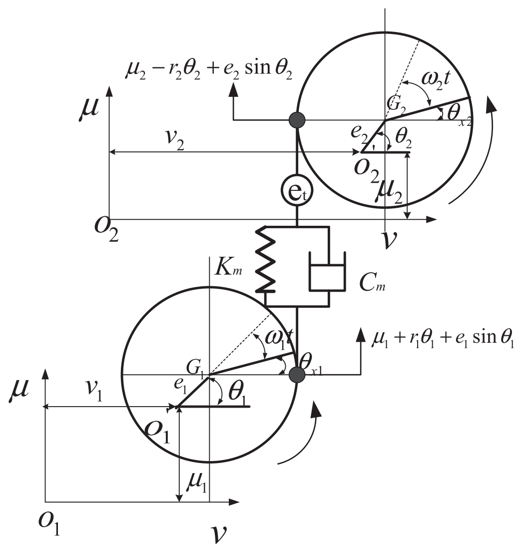


Fig. 11. Typical dynamic model of gear pair.

$$J_{d1}\ddot{\theta}_{\nu 1} + J_{p1}\omega_1\dot{\theta}_{\mu 1} = 0 \tag{37}$$

$$J_{d1}\ddot{\theta}_{\mu 1} - J_{p1}\omega_1\dot{\theta}_{\nu 1} = 0 \tag{38}$$

$$\begin{aligned}
&J_{p1}\ddot{\theta}_1 + K_m(r_1\mu_1 + r_1^2\theta_1 - r_1\mu_2 + r_1r_2\theta_2) + C_m\left(r_1\dot{\mu}_1 + r_1^2\dot{\theta}_1 - r_1\dot{\mu}_2 + r_1r_2\dot{\theta}_2\right) = \\
&\quad\Big[-K_m(\mu_1 + r_1\theta_1 - \mu_2 + r_2\theta_2) - C_m\left(\dot{\mu}_1 + r_1\dot{\theta}_1 - \dot{\mu}_2 + r_2\dot{\theta}_2\right) + K_m(e_2\sin\theta_2 - e_1\sin\theta_1 + e_t) + \\
&\quad C_m\left(e_2\omega_2\sin\theta_2 - e_1\omega_1\sin\theta_1 + \dot{e}_t\right)\Big]e_1\cos\theta_1 + \Big[K_m(e_2\sin\theta_2 - e_1\sin\theta_1 + e_t) + C_m(e_2\omega_2\sin\theta_2 - e_1\omega_1\sin\theta_1 + \dot{e}_t)\Big]r_1
\end{aligned} \tag{39}$$

$$J_{d2}\ddot{\theta}_{\nu 2} + J_{p2}\omega_2\dot{\theta}_{\mu 2} = 0 \tag{40}$$

$$J_{d2}\ddot{\theta}_{u2} - J_{p2}\omega_2\dot{\theta}_{\nu 2} = 0 \tag{41}$$

$$\begin{aligned}
&J_{p2}\dot{\theta}_2 + K_m(r_2\mu_1 + r_1r_2\theta_1 - r_2\mu_2 + r_2^2\theta_2) + C_m(r_2\dot{\mu}_1 + r_1r_2\dot{\theta}_1 - r_2\dot{\mu}_2 + r_2^2\dot{\theta}_2) = \\
&\quad\Big[-K_m(\mu_1 + r_1\theta_1 - \mu_2 + r_2\theta_2) - C_m(\dot{\mu}_1 + r_1\dot{\theta}_1 - \dot{\mu}_2 + r_2\dot{\theta}_2) + K_m(e_2\sin\theta_2 - e_1\sin\theta_1 + e_t) + \\
&\quad C_m\left(e_2\omega_2\sin\theta_2 - e_1\omega_1\sin\theta_1 + \dot{e}_t\right)\Big]e_2\cos\theta_2 + \Big[K_m(e_2\sin\theta_2 - e_1\sin\theta_1 + e_t) + C_m(e_2\omega_2\sin\theta_2 - e_1\omega_1\sin\theta_1 + \dot{e}_t)\Big]r_2
\end{aligned} \tag{42}$$

where M_1 and M_2 represent the masses, J_d and J_p are the transverse and polar moments of inertia, K_m and C_m are the gear time varying mesh stiffness and damping, r_1 and r_2 are the gear base circles, e_1 and e_2 are the geometrical eccentricities, ω_1 and ω_2 are the rotation speeds, θ_1 and θ_2 are the rotation angles, respectively. e_t represents GTE.

The coupled equation of motion of a gear pair by arranging them in a matrix form can be given by

$$\mathbf{M}_G\ddot{\mathbf{q}}_G + (\mathbf{G}_G + \mathbf{C}_G)\dot{\mathbf{q}}_G + \mathbf{K}_G\mathbf{q}_G = \mathbf{Q}_G \tag{43}$$

where $\mathbf{M}_G$, $\mathbf{K}_G$, $\mathbf{G}_G$, $\mathbf{C}_G$ are the inertia, damping, gyroscopic and stiffness matrices, respectively. $\mathbf{Q}_G$ is the external force vector in the local pressure line co-ordinate system. They are given in Appendix B by Eqs. (B.1)–(B.14).

3.2.2. Geometric transmission error

Transmission Error (TE) is one of the most important and fundamental concepts that forms the basis of understanding vibrations in gears. It was originally coined by Professor S.L. Harris from Lancaster University. TE is defined as the deviation of the angular position of the driven gear from its theoretical position calculated from the gearing ratio and the angular position of the pinion. TE was produced by many sources, such as gear case accuracy, pinion and gear profile accuracy, and pinion and gear Tooth deflection. The GTE e_t shown in formula (33)–(42) is one form of TE.

A typical GTE of a spur gear is shown in Fig. 12. It shows a long periodic wave (gear shaft rotation) and short regular waves occurring at tooth-mesh frequency. The long wave is often known as Long Term Component (LTC) which is typically caused by the eccentricity of the gear about its rotational center, while the short waves are known as Short Term Component (STC) mainly caused by gear tooth profile errors and base pitch spacing error between the teeth. In this model the GTE e_t was estimated based on combined waveform expression of eccentricity and tooth-profile errors as follows:

$$e_t = e_{eccentricities} + e_{tooth\text{-}profile} \tag{44}$$

$$e_{eccentricities} = e_1\sin(2\pi f_{n1}t + \varphi_1) + e_2\sin(2\pi f_{n2}t + \varphi_2) \tag{45}$$

$$e_{tooth\text{-}profile} = A\sin(2\pi f_p t + \varphi) \tag{46}$$

where e_1 and e_2 are the eccentricity of driving and driven gears, f_{n1} and f_{n2} are shaft frequency of driving and driven gears, f_p is meshing frequency, φ_1, φ_2, φ are the phase difference, A is the amplitude of tooth-profile error.

3.2.3. The assembly of system dynamic equations

The total stiffness matrix $\mathbf{K}_{total}$ of gear-rotor system can be expressed as shown in Fig. 13. $\square$ Indicates the total mass matrix of rotor 1, $\square$ indicates the total mass matrix of rotor 2. $\mathbf{K}_G^{11}$, $\mathbf{K}_G^{12}$, $\mathbf{K}_G^{21}$, $\mathbf{K}_G^{22}$ are block matrix of gear pair mass matrix $\mathbf{K}_G$, they can be assembled on to the corresponding node of rotor 1 and rotor 2. By using the same method, the total gyroscopic matrix $\mathbf{G}_{total}$, mass matrix $\mathbf{M}_{total}$ and damping matrix $\mathbf{C}_{total}$ also can be obtained. Then the system equations of motion can be expressed as

$$\mathbf{M}_{total}\ddot{\mathbf{q}} + (\mathbf{G}_{total} + \mathbf{C}_{total})\dot{\mathbf{q}} + \mathbf{K}_{total}\mathbf{q} = \mathbf{Q}_{total} \tag{47}$$

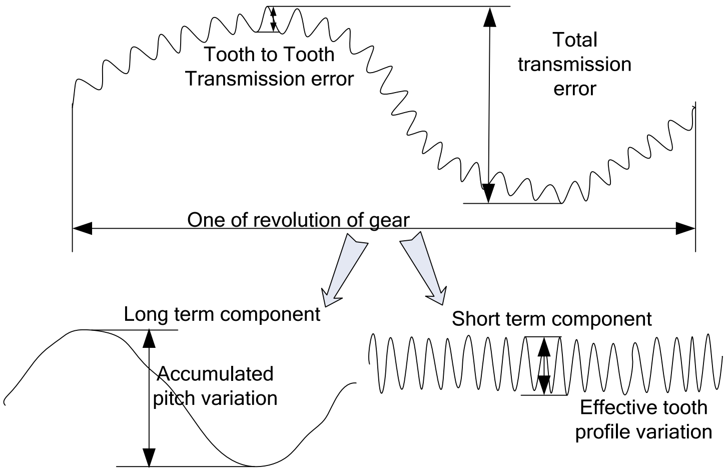


Fig. 12. A typical geometrical transmission error.

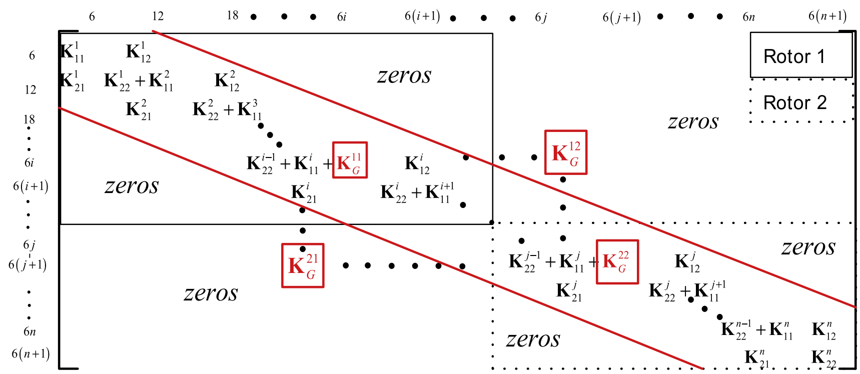


Fig. 13. Total stiffness matrix of gear-rotor system.

4. Simulation and experimental results and discussion

4.1. Simulation results

4.1.1. Dynamic simulation of gear-rotor system with tooth root crack

In order to compare the simulation results with those obtained experimentally, the simulation parameters is same to the test rig parameters which are listed in Table 5. The mesh stiffness calculated by the mathematical model discussed in Section 2 is plugged into the dynamic Eq. (47), then use the integral calculus method of Newmark-β to solve the equation, and get each node and element's displacement and acceleration versus time. The results are presented in Fig. 14.

As shown in Fig. 14, the simulated signal shows that an impulse period caused by tooth crack is $\Delta_t = 0.190$ s$\left(f = \frac{1}{\Delta_t} = 5.23\ \text{Hz}\right)$. This period is equal to the rotational period of the defected gear. Fig. 15 (partial enlarged view of Fig. 14) shows that the dynamic response of the gear-rotor system also is periodic impulses during meshing of every gear tooth. The frequency (f_m = 394 Hz) of every two periodic impulses is equal to mesh frequency. It can be concluded that the response of spur gear-rotor system is periodic impulses due to the periodic sudden change of time varying mesh stiffness. When the cracked tooth comes in contact, the impulse amplitude will increase as a result of reductions of mesh stiffness. The larger amplitude impulses contain useful diagnostic information which is important for extracting features of tooth defects.

4.1.2. Dynamic analysis of gear-rotor system with tooth root crack

4.1.2.1. The effects of GTE on vibration response. Fig. 16 is the response of the dynamic model with different GTE. In Fig. 16(a), the eccentricity of the pinion and gear is 200×10^{-6} m, and the amplitude of tooth profile error is 30×10^{-6} m. In Fig. 16(b), the parameters of GTE are zero. As shown in Fig. 16, the simulated results show that the transmission error will produce amplitude modulation phenomenon. Fig. 17 is detail view of Fig. 16. By comparing Fig. 17(a) and (b), we can find a sinusoidal frequency components caused by the Short Term Component (STC) of GTE. Through involving the GTE will cause some

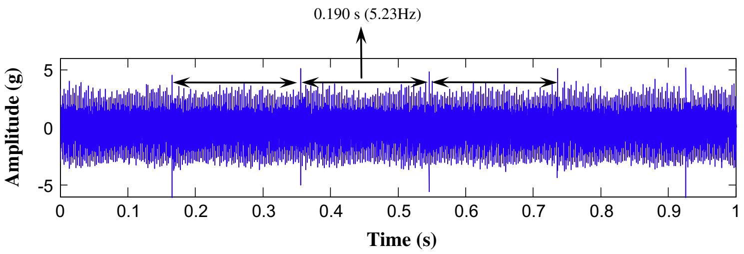


Fig. 14. The simulation result in y direction of bearing.

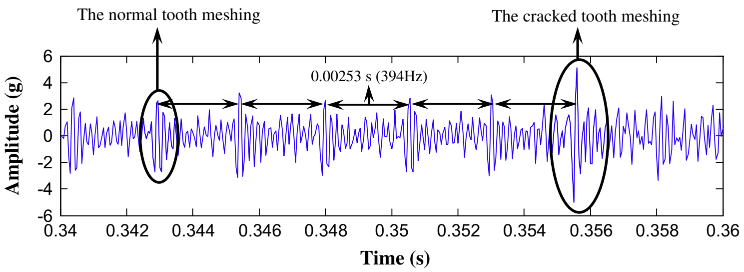


Fig. 15. Enlarged view of Fig. 14.

changes mentioned above, the periodic impulses caused by the cracked tooth also are very clear. Through the above analysis, we can concluded that involve the GTE bring difficulties to the diagnosis of the gear tooth root crack in some degree, but the fault characteristics also can be found.

4.1.2.2. The effects of bearing stiffness on dynamic characteristics. The amplitude–frequency response curves of the first natural frequency at the bearing of driving shaft are shown in Fig. 18 when the bearing stiffness are 1×10^7, 5×10^7, 1×10^8 N/m. It shows that the natural frequency becomes larger and the peak values decrease as the bearing stiffness increases. Fig. 19 is the vibration response when the bearing stiffness are1×10^8, 5×10^7 N/m. It is found that the amplitude of the vibration response decreases from 5 g to 7 g when the bearing stiffness increases from 5×10^7 N/m to 1×10^8 N/m. The periodic impulses caused by the cracked tooth also are very clear.

4.1.2.3. The effects of gear mesh stiffness on dynamic characteristics. Fig. 20 shows the coupled lateral and torsional natural frequencies versus gear mesh stiffness. It is found that as gear mesh stiffness K_m increases, some frequencies increases greatly within certain limits and thereafter reaches a constant, while others nearly remain constant. The former represent the lateral–torsional coupling frequency and the later represent the lateral natural frequency.

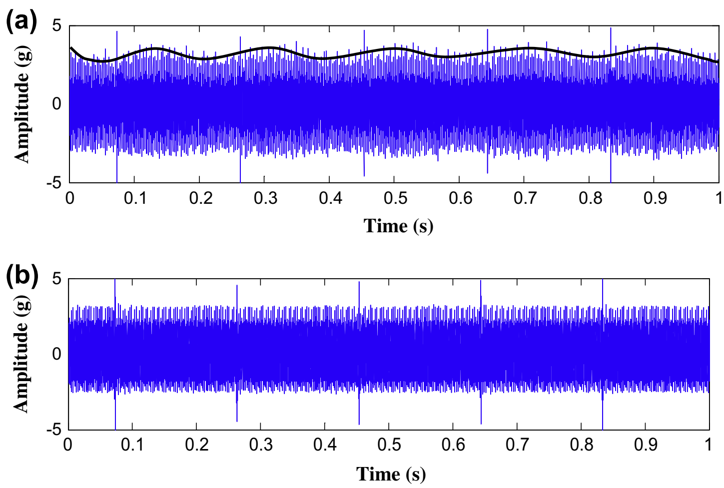


Fig. 16. The simulation signal of dynamic model with different geometric transmission error. (a) $e_1 = e_2 = 200 \times 10^{-6}$ m, $A = 30 \times 10^{-6}$ m. (b) $e_1 = e_2 = 0$ mm, $A = 0$ m.

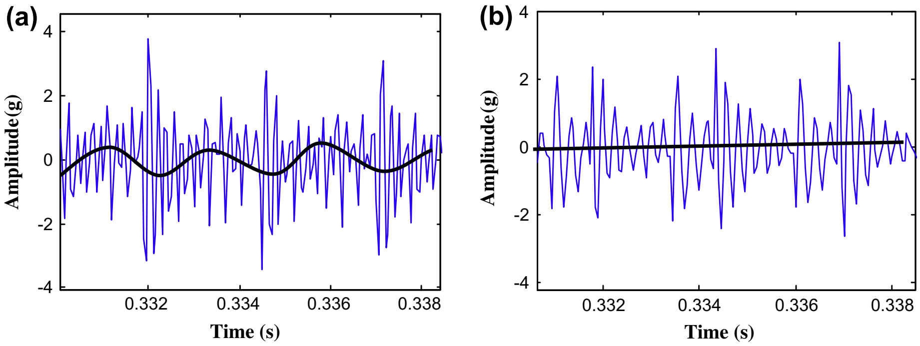


Fig. 17. Detail view of Fig. 16.

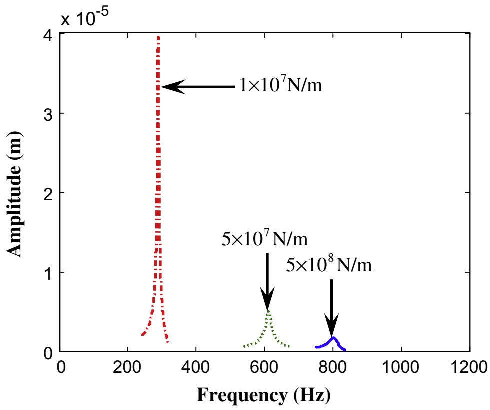


Fig. 18. The amplitude–frequency response curves of the first natural frequency with different bearing stiffness.

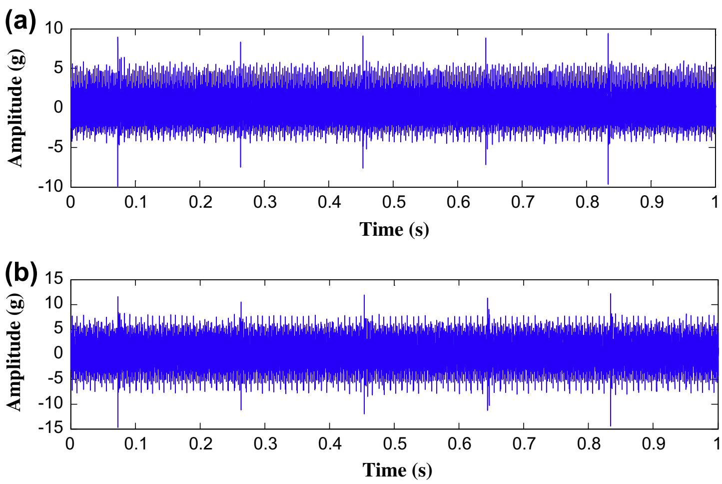


Fig. 19. the vibration response when the bearing stiffness is (a) 1×10^8 N/m and (b) 5×10^7 N/m.

The vibration responses when the mean value of mesh stiffness are 3.5×10^7, 3.5×10^8 N/m are shown in Fig. 21. It is found that the amplitude of the vibration response decreases when the mesh stiffness increase from 5×10^7 N/m to 1×10^8 N/m. Fig. 22 is detail view of Fig. 21. By comparing Fig. 22(a) and (b), it can be found the frequency of impulses decreases as the mesh stiffness increases.

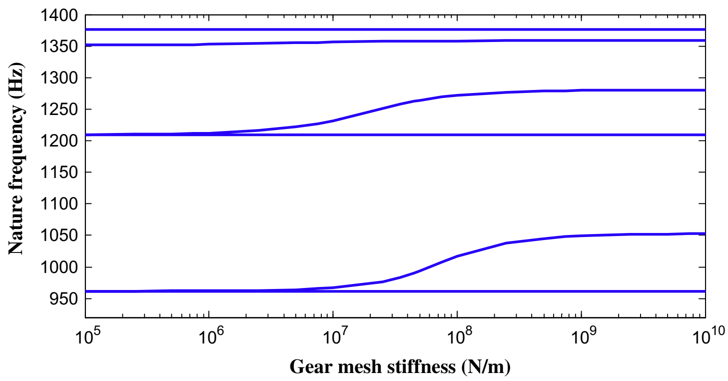


Fig. 20. Natural frequencies versus gear mesh stiffness.

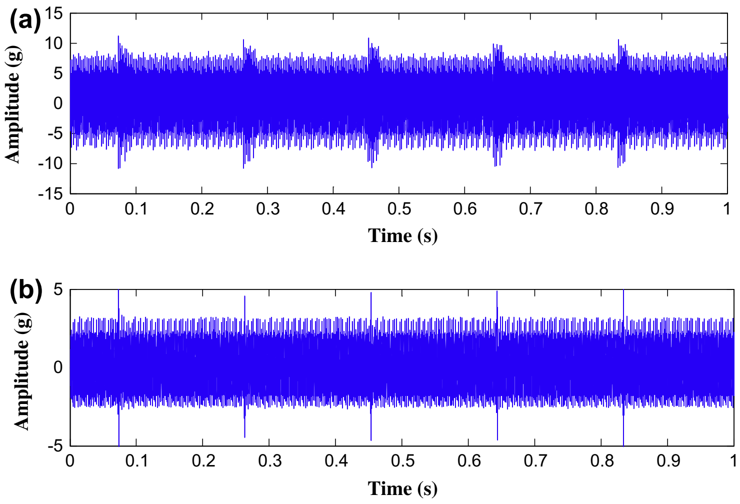


Fig. 21. The vibration response when the mean value of mesh stiffness is (a) 3.5×10^7 N/m and (b) 3.5×10^8 N/m.

4.2. Experimental results

A photograph of the spur gear rig is shown in Fig. 23(a) and the tested gear with tooth crack is shown in Fig. 23(b). The parameters of this test rig are shown in Table 5. The dimensions of the crack fault are 2 mm depth and 20 mm length across the tooth root (100% tooth root width).The vibration acceleration signals of the gear system under fault states can be collected using an accelerometer positioned on the top of the gearbox casing, the sampling frequency is 12.8 kHz, the rotational frequency of the driving gear f_{p1} is about 7.2 Hz, the rotational frequency of the driving gear f_{p2} is about 5.26 Hz, the mesh frequency is 394 Hz.

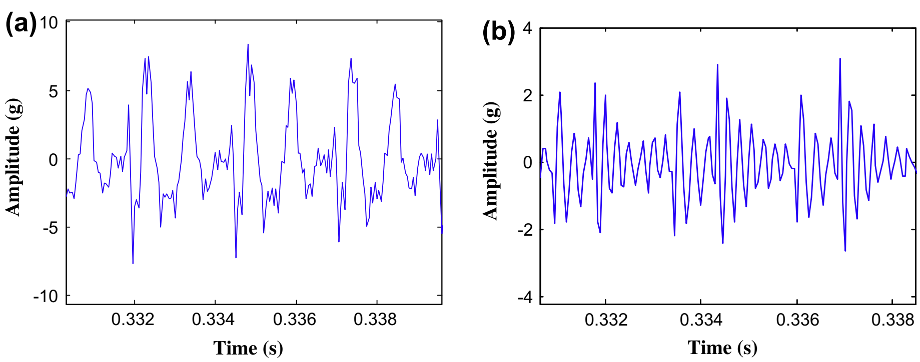


Fig. 22. Detail view of Fig. 21.

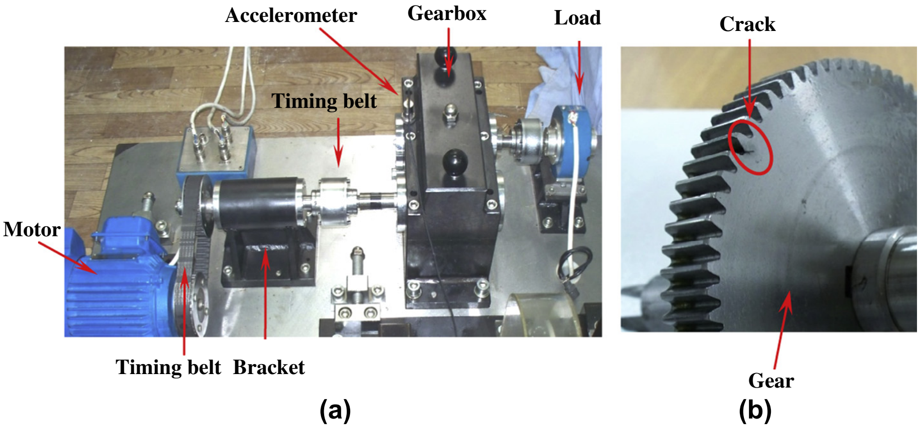


Fig. 23. (a) The spur gear rig. (b) The gear with tooth crack.

The measured signal and its spectrum of the spur gear rig are shown in Fig. 24. 394 Hz, 788 Hz, 1182 Hz is the mesh frequency and its harmonic components. The periodic impulses stemmed from the crack fault should occur, but they are drowned in heavy noise, the incipient fault signatures hidden in gearbox signals are always too weak to be identified, evident sidebands relevant to the crack fault can rarely be found in Fig. 24(b). The redundant lifting scheme (second wavelet transform) is well known for its ability to extract transient features. The shape of the wavelet function is very similar to an impulse. Therefore, it is desirable to extract the transient components from vibration signals. An improved redundant second generation wavelet transforms was presented by Li et al. [24], this method can be introduced briefly as follows.

Suppose the original signal is $\mathbf{s}^{(l)}$. The forward transform of the improved redundant second generation wavelet transforms is shown in Fig. 25. The improved redundant lifting scheme consists in three main steps:

$$(1)\ \text{Prediction}:\ \mathbf{d}^{(l+1)} = \mathbf{s}^{(l)} - P^{(l)}\mathbf{s}^{(l)}. \tag{48}$$

$$(2)\ \text{Update}:\ \mathbf{s}^{(l+1)} = \mathbf{s}^{(l)} + U^{(l)}\mathbf{d}^{(l+1)}. \tag{49}$$

$$(3)\ \text{Normalization}:\ \mathbf{r}^{(l+1)} = b_{l+1}\mathbf{d}^{(l+1)}, \mathbf{c}^{(l+1)} = a_{l+1}\mathbf{s}^{(l+1)}. \tag{50}$$

where $P^{(l)}$, $U^{(l)}$ are the redundant prediction operator and the redundant update operator, respectively. The $\mathbf{d}^{(l+1)}$ and $\mathbf{s}^{(l+1)}$ are computed by the classical redundant lifting scheme. $\mathbf{r}^{(l+1)}$ and $\mathbf{c}^{(l+1)}$ are detail signal and approximation signal, the a_{l+1} and b_{l+1} are the normalization factors at level $l + 1$.

As shown in Fig. 25, by using the improved redundant second generation wavelet transform, the original signal is decomposed into two subbands, namely the approximation coefficients and detail coefficients. Three-level decomposition by applying this method to decompose, the original signal obtained 8 subbands, and the first 4 subbands are illustrated in Fig. 26.

As shown in Fig. 26, the fault characteristics of the first 4 subspace are the most obviously. The partial detail view of the fourth subband is shown in Fig. 27, it shows that the gear-rotor system generate periodic impulse whose frequency is mesh

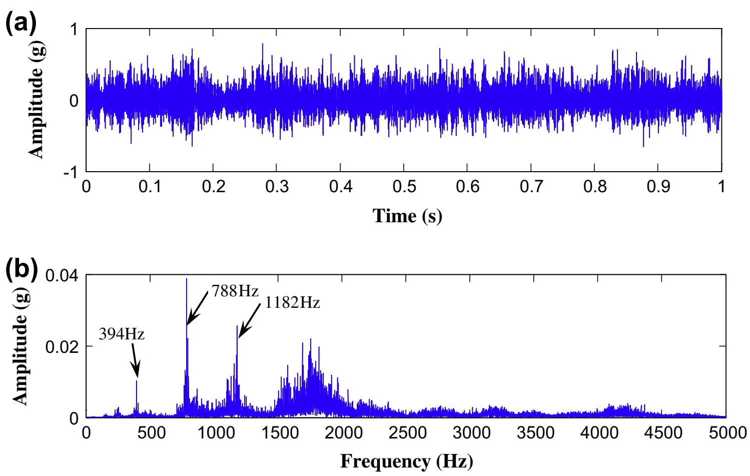


Fig. 24. (a) The vibration signal of the gearbox and (b) the FFT spectrum.

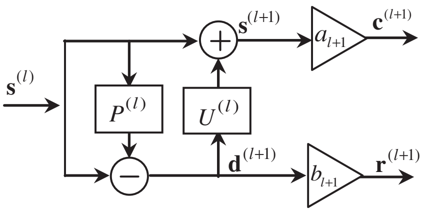


Fig. 25. The forward transform of improved redundant lifting scheme.

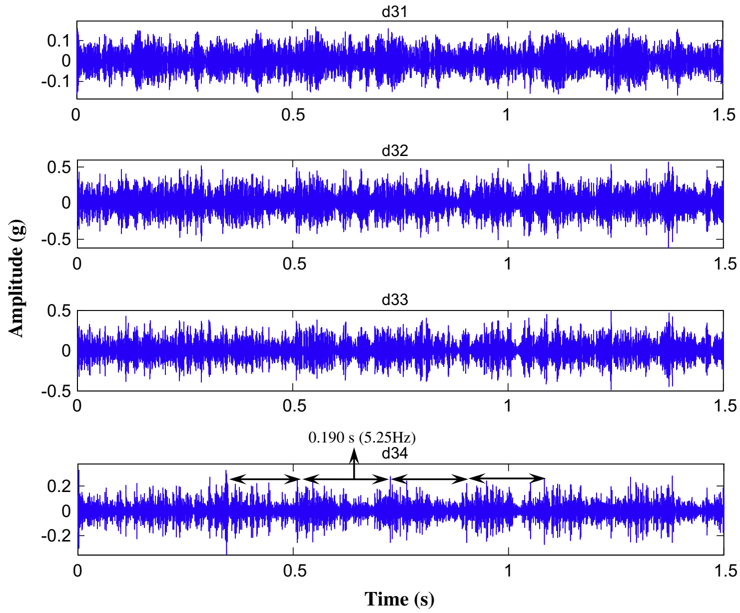


Fig. 26. The first 4 subspaces of original signal.

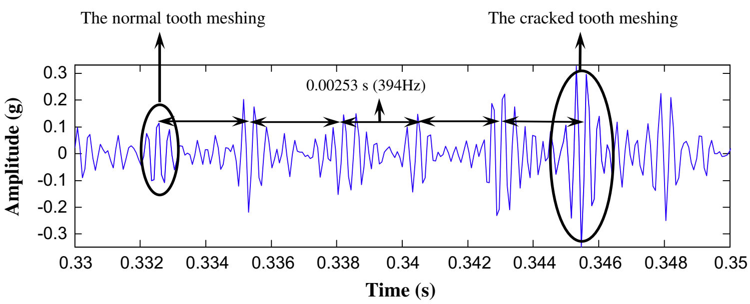


Fig. 27. Enlarged view of Fig. 26.

frequency f_m = 394 Hz, when the crack tooth comes into contact, the impulse amplitude will increase. Results of simulation agree well with those obtained by experiments.

5. Conclusions

To ascertain the effect of gear crack damage on the gear case vibration, this paper through computer simulation studies the effects of tooth crack growth on the vibration response of gear-rotor system with spur gears. The following conclusions can be obtained:

(1) The gear vibration signals vary throughout one complete mesh cycle because of variations in gear meshing stiffness, transmission error.
(2) When calculating the mesh stiffness by the so-called potential energy method, the flexibility between the root circle and base circle should be considered.
(3) When the cracked tooth on gear comes in contact during meshing, the mesh stiffness reduces. The total effective mesh stiffness reduces further as crack length increases. Due to the periodic sudden change of time varying mesh stiffness, the dynamic response of the gear-rotor system is periodic impulses. When the cracked tooth comes in contact, the impulse amplitude will increase as a result of reductions of mesh stiffness. The influence caused by the cracked tooth repeats only once in a revolution, the duration between every two larger amplitude impulses is equal to rotational frequency of the defected gear.
(4) Amplitude modulation phenomenon due to the Long Term Component (LTC) of GTE and a sinusoidal frequency components which is caused by the Short Term Component (STC) of GTE can both be found in the simulated signal. The natural frequency enlarges as the bearing stiffness increases. The lateral–torsional coupling frequency increases greatly within certain limits and thereafter reaches a constant while the lateral natural frequency nearly remains constant as the gear mesh stiffness increases. The amplitude of the simulated signal decreases as the mesh stiffness or bearing stiffness increase.

Dynamic modeling can improve the understanding of the diagnostic information that buried in the vibration signal. But theoretical model is much different from test bench. There are some disparities between the simulation results and experiment results. Future research should consider more influences on the vibration response, such as friction and inter-tooth backlash in meshing, and manufacturing errors in gears.

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Appendix A

$$\mathbf{M}_T^e = \frac{\rho A l}{420}\begin{bmatrix}
140 & 0 & 0 & 0 & 0 & 0 & 70 & 0 & 0 & 0 & 0 & 0 \\
 & 156 & 0 & 0 & 0 & 22l & 0 & 54 & 0 & 0 & 0 & -13l \\
 & & 156 & 0 & -22l & 0 & 0 & 0 & 54 & 0 & 13l & 0 \\
 & & & \frac{140}{A}J & 0 & 0 & 0 & 0 & 0 & \frac{70}{A}J & 0 & 0 \\
 & & & & 4l^2 & 0 & 0 & 0 & -13l & 0 & -3l^2 & 0 \\
 & & & & & 4l^2 & 0 & 13l & 0 & 0 & 0 & -3l^2 \\
 & & & & & & 140 & 0 & 0 & 0 & 0 & 0 \\
 & & & & & & & 156 & 0 & 0 & 0 & -22l \\
 & & & sym & & & & & 156 & 0 & 22l & 0 \\
 & & & & & & & & & \frac{140}{A}J & 0 & 0 \\
 & & & & & & & & & & 4l^2 & 0 \\
 & & & & & & & & & & & 4l^2
\end{bmatrix} \tag{A.1}$$

$$\mathbf{M}_R^e = \frac{\rho A r^2}{120 l}\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 & 36 & 0 & 0 & 0 & 3l & 0 & -36 & 0 & 0 & 0 & 3l \\
 & & 36 & 0 & -3l & 0 & 0 & 0 & -36 & 0 & -3l & 0 \\
 & & & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 & & & & 4l^2 & 0 & 0 & 0 & 3l & 0 & -l^2 & 0 \\
 & & & & & 4l^2 & 0 & -3l & 0 & 0 & 0 & -l^2 \\
 & & & & & & 0 & 0 & 0 & 0 & 0 & 0 \\
 & & & & & & & 36 & 0 & 0 & 0 & -3l \\
 & & & sym & & & & & 36 & 0 & 3l & 0 \\
 & & & & & & & & & 0 & 0 & 0 \\
 & & & & & & & & & & 4l^2 & 0 \\
 & & & & & & & & & & & 4l^2
\end{bmatrix} \tag{A.2}$$

$$\mathbf{G}^e = \frac{\rho A r^2}{60l}\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & 0 & -36 & 0 & 3l & 0 & 0 & 0 & 36 & 0 & 3l & 0 \\ & & 0 & 0 & 0 & 3l & 0 & -36 & 0 & 0 & 0 & 3l \\ & & & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & & 0 & -4l^2 & 0 & 3l & 0 & 0 & 0 & l^2 \\ & & & & & 0 & 0 & 0 & 3l & 0 & -l^2 & 0 \\ & & & & & & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & & & & & 0 & -36 & 0 & -3l & 0 \\ & & \mathit{skew} & \mathit{sym} & & & & & 0 & 0 & 0 & -3l \\ & & & & & & & & & 0 & 0 & 0 \\ & & & & & & & & & & 0 & -4l^2 \\ & & & & & & & & & & & 0 \end{bmatrix} \tag{A.3}$$

$$\mathbf{K}^e = \begin{bmatrix} \frac{EA}{l} & 0 & 0 & 0 & 0 & 0 & -\frac{EA}{l} & 0 & 0 & 0 & 0 & 0 \\ & \frac{12EI_z}{l^3} & 0 & 0 & 0 & \frac{6EI_z}{l^2} & 0 & -\frac{12EI_z}{l^3} & 0 & 0 & 0 & \frac{6EI_z}{l^2} \\ & & \frac{12EI_y}{l^3} & 0 & -\frac{6EI_y}{l^2} & 0 & 0 & 0 & -\frac{12EI_y}{l^3} & 0 & -\frac{6EI_y}{l^2} & 0 \\ & & & \frac{GJ}{l} & 0 & 0 & 0 & 0 & 0 & -\frac{GJ}{l} & 0 & 0 \\ & & & & \frac{4EI_y}{l} & 0 & 0 & 0 & \frac{6EI_y}{l^2} & 0 & \frac{2EI_y}{l} & 0 \\ & & & & & \frac{4EI_z}{l} & 0 & -\frac{6EI_z}{l^2} & 0 & 0 & 0 & \frac{2EI_z}{l} \\ & & & & & & \frac{EA}{l} & 0 & 0 & 0 & 0 & 0 \\ & & & & & & & \frac{12EI_z}{l^3} & 0 & 0 & 0 & -\frac{6EI_z}{l^2} \\ & & & \mathit{sym} & & & & & \frac{12EI_y}{l^3} & 0 & \frac{6EI_y}{l^2} & 0 \\ & & & & & & & & & -\frac{GJ}{l} & 0 & 0 \\ & & & & & & & & & & \frac{4EI_y}{l} & 0 \\ & & & & & & & & & & & \frac{4EI_z}{l} \end{bmatrix} \tag{A.4}$$

where ρ, A, l, r are the density, cross-sectional area length and radius of axis element, respectively. G, E, J represent shear modulus, Young's modulus, polar moments of inertia, respectively.

Appendix B

$$\mathbf{K}^G = \begin{bmatrix} \mathbf{K}_G^{11} & \mathbf{K}_G^{12} \\ \mathbf{K}_G^{21} & \mathbf{K}_G^{22} \end{bmatrix} \tag{B.1}$$

where

$$\mathbf{K}_G^{11} = K_m\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & r_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & r_1 & 0 & r_1^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{K}_G^{12} = K_m\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & r_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -r_1 & 0 & r_1 r_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{B.2}$$

$$\mathbf{K}_G^{21} = K_m\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -r_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & r_2 & 0 & r_1 r_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{K}_G^{22} = K_m\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -r_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -r_2 & 0 & r_2^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{B.3}$$

$$\mathbf{C}^G = \begin{bmatrix} \mathbf{C}_G^{11} & \mathbf{C}_G^{12} \\ \mathbf{C}_G^{21} & \mathbf{C}_G^{22} \end{bmatrix} \tag{B.4}$$

where

$$\mathbf{C}_G^{11} = C_m \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & r_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & r_1 & 0 & r_1^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{C}_G^{12} = C_m \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -r_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -r_1 & 0 & r_1 r_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{B.5}$$

$$\mathbf{C}_G^{21} = C_m \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -r_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & r_2 & 0 & r_1 r_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{C}_G^{22} = C_m \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & r_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -r_2 & 0 & r_2^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{B.6}$$

$$\mathbf{M}_G = \begin{bmatrix} \mathbf{M}_G^{11} & 0 \\ 0 & \mathbf{M}_G^{22} \end{bmatrix} \tag{B.7}$$

where

$$\mathbf{M}_G^{11} = \begin{bmatrix} m_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & m_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & m_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & J_{p1} & 0 & 0 \\ 0 & 0 & 0 & 0 & J_{d1} & 0 \\ 0 & 0 & 0 & 0 & 0 & J_{d1} \end{bmatrix}, \quad \mathbf{M}_G^{11} = \begin{bmatrix} m_2 & 0 & 0 & 0 & 0 & 0 \\ 0 & m_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & m_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & J_{p2} & 0 & 0 \\ 0 & 0 & 0 & 0 & J_{d2} & 0 \\ 0 & 0 & 0 & 0 & 0 & J_{d2} \end{bmatrix} \tag{B.8}$$

$$\mathbf{G}_G = \begin{bmatrix} \mathbf{G}_G^{11} & 0 \\ 0 & \mathbf{G}_G^{22} \end{bmatrix} \tag{B.9}$$

where

$$\mathbf{G}_G^{11} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -J_{p1}\omega_1 \\ 0 & 0 & 0 & 0 & J_{p1}\omega_1 & 0 \end{bmatrix}, \quad \mathbf{G}_G^{11} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -J_{p2}\omega_2 \\ 0 & 0 & 0 & 0 & J_{p2}\omega_2 & 0 \end{bmatrix} \tag{B.10}$$

$$\mathbf{Q}_G = \begin{bmatrix} \mathbf{Q}_G^1 \\ \mathbf{Q}_G^2 \end{bmatrix} \tag{B.11}$$

where

$$\mathbf{Q}_G^1 = \begin{bmatrix} 0 \\ m_1 e_1 \omega_1^2 \sin\theta_1 + F_1 \\ m_1 e_1 \omega_1^2 \cos\theta_1 \\ F_s e_1 \cos\theta_1 - F_1 r_1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{Q}_G^2 = \begin{bmatrix} 0 \\ m_2 e_2 \omega_2^2 \sin\theta_2 - F_1 \\ m_2 e_2 \omega_2^2 \cos\theta_2 \\ F_s e_2 \cos\theta_2 - F_1 r_2 \\ 0 \\ 0 \end{bmatrix} \tag{B.12}$$

$$F_1 = K_m(e_2 \sin\theta_2 - e_1 \sin\theta_1 + e_t) + C_m(e_2\omega_2 \sin\theta_2 - e_1\omega_1 \sin\theta_1 + \dot{e}_t) \tag{B.13}$$

$$F_s = -K_m(\mu_1 + r_1\theta_1 - \mu_2 + r_2\theta_2) - C_m\left(\dot{\mu}_1 + r_1\dot{\theta}_1 - \dot{\mu}_2 + r_2\dot{\theta}_2\right) + F_1 \tag{B.14}$$

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